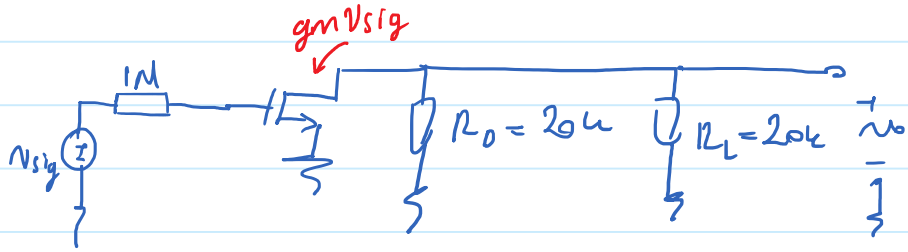
Q1

7.64 Calculate the overall voltage gain of a CS amplifier fed with a 1- MΩ source and connected to a 20- kΩ load. The MOSFET has $g_m = 2$ mA/V, and a drain resistance $R_D = 20$ kΩ is utilized.



$$G_v = \frac{V_o}{V_{sig}} = (20k // 20k)\ 2m = -20\ V/V$$

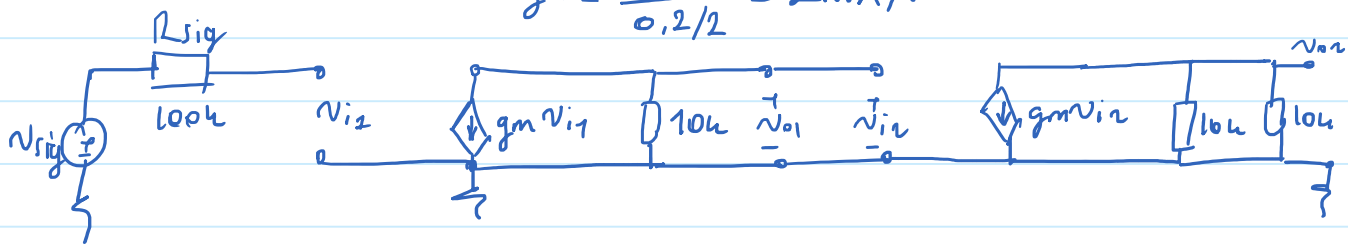
Q2

7.67 Two identical CS amplifiers are connected in cascade. The first stage is fed with a source v_{sig} having a resistance R_{sig} = 100 kΩ. A load resistance R_L = 10 kΩ is connected to the drain of the second stage. Each MOSFET is biased at I_D = 0.2 mA and operates with V_{OV} = 0.2 V. Each stage utilizes a drain resistance R_D = 10 kΩ.

(a) Sketch the equivalent circuit of the two-stage amplifier.

(b) Calculate the overall voltage gain G_v.

$$g_m = \frac{0.2m}{0.2/2} = 2mA/V$$

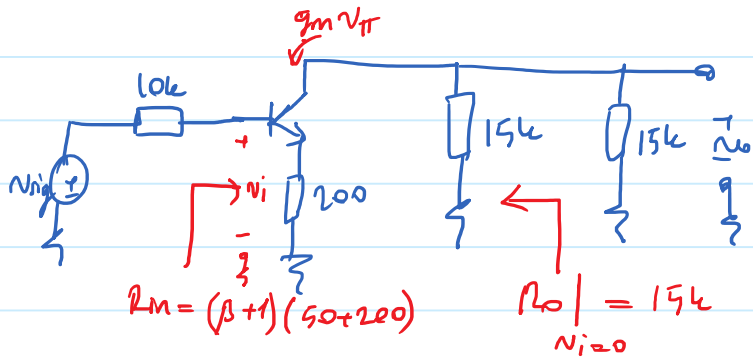


$$G_v = \frac{v_{o2}}{v_{sig}} = \frac{v_{i1}}{v_{sig}} \cdot \frac{v_{o1}}{v_{i1}} \cdot \frac{v_{i2}}{v_{o1}} \cdot \frac{v_{o2}}{v_{i2}}$$

$$= (-g_m \cdot 10k) \cdot (-g_m \, 5k) = 20 \cdot 10 = 200 \, V/V$$

Q3

7.74 A CE amplifier uses a BJT with $\beta = 100$ biased at $I_C = 0.5$ mA and has a collector resistance $R_C = 15$ kΩ and a resistance R_e =200Ω connected in the emitter. Find R_{in}, A_{vo}, and R_o. If the amplifier is fed with a signal source having a resistance of 10 kΩ, and a load resistance R_L = 15 kΩ is connected to the output terminal, find the resulting A_v and G_v. If the peak voltage of the sine wave appearing between base and emitter is to be limited to 5 mV, what $\hat{v}_{sig}$ is allowed, and what output voltage signal appears across the load?



$g_m = \frac{0.5m}{0.025} = 20\ mA/V$

$r_e \cong 1/g_m = 50\ \Omega$

$$V_\pi = \frac{V_i}{200+50} \cdot 50 = \frac{V_i}{5}$$

$$V_o = -g_m V_\pi \cdot 15k = -20m \cdot V_\pi \cdot 15k \quad \text{(without } R_L\text{)}$$

$$V_o = -300 \frac{V_i}{5} = -60 V_i$$

$$A_{vo} = -60\ V/V$$

$$R_{in} = (\beta+1)(50+200) = 101 \cdot (250) \cong 25k\Omega$$

$$R_o = 15k\Omega$$

$$A_v = \frac{A_{vo} R_L}{R_o + R_L} = \frac{-60 \cdot 15k}{15k+15k} = -30\ V/V$$

$$G_v = \frac{V_i}{V_{sig}} \cdot \underbrace{\frac{V_o}{V_i}}_{A_v} = \frac{25k}{25k+10k} \cdot -30 = \frac{-150}{7}\ V/V$$

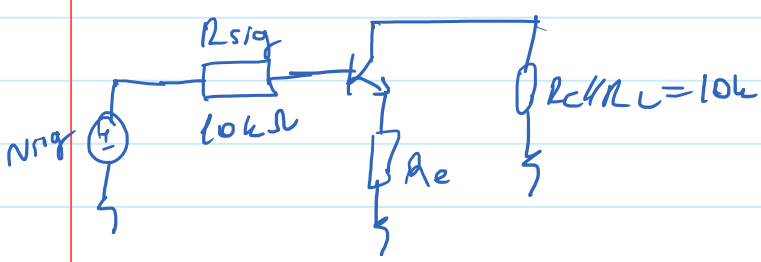
$$V_{\pi max} = 5mV = \frac{V_{imax}}{5} \Rightarrow V_{imax} = 25mV$$

$$V_{sigmax} = V_{imax} \cdot \frac{25k}{35k} = 35mV \Rightarrow V_{omax} = G_v \cdot V_{sigmax}$$

$$V_{omax} = \frac{-150}{7} \cdot 35m = -750mV$$

Q4

SIM **D 7.76** Including an emitter resistance R_e reduces the variability of the gain G_v due to the inevitable wide variance in the value of β. Consider a CE amplifier operating between a signal source with R_{sig} = 10 kΩ and a total collector resistance $R_C \parallel R_L$ of 10 kΩ. The BJT is biased at I_C = 1 mA and its β is specified to be nominally 100 but can lie in the range of 50 to 150. First determine the nominal value and the range of $|G_v|$ without resistance R_e. Then select a value for R_e that will ensure that $|G_v|$ be within ±20% of its new nominal value. Specify the value of R_e, the new nominal value of $|G_v|$, and the expected range of $|G_v|$.



$$g_m = \frac{I_C}{V_T} = \frac{1m}{0.025V} = 40\,mA/V$$

$$\beta_{nom} = 100, \quad 50 < \beta < 150$$

$$r_{\pi_{nom}} = \frac{V_T}{I_B} = \frac{25m}{1m/100} = 2.5k\Omega$$

$$1.25k < r_\pi < 3.75k$$

Without Re : $G_v = -\frac{r_\pi}{r_\pi + 10k}\cdot\frac{\beta}{r_e}\cdot(R_C \parallel R_L)$

$$G_v \cong -\beta\,\frac{R_C \parallel R_L}{r_\pi + 10k}$$

β	50	100	150
$\lvert G_v \rvert$	44	80	109

with Re

$$G_v = -\beta\,\frac{R_C \parallel R_L}{(\beta+1)(r_e+R_e)+10k}$$

$\beta = 50$ → $G_{v_{min}} \cong -50\,\frac{10k}{1.25k + 51R_e + 10k}$

$\beta = 100$ → $G_{v_{nom}} \cong -100\,\frac{10k}{2.5k + 100R_e + 10k}$

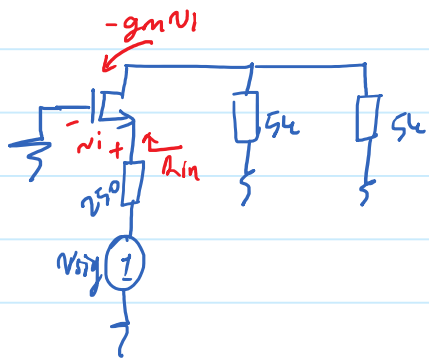
$$\frac{G_{v_{min}}}{G_{v_{nom}}} = 0.8$$

$$\frac{1}{2}\,\frac{12.5k + 100R_e}{11.25k + 100R_e} = 0.8 \Rightarrow R_e \cong 28.4\Omega$$

...

Q5

7.77 A CG amplifier using an NMOS transistor for which g_m =5 mA/V has a 5- kΩ drain resistance R_D and a 5- kΩ load resistance R_L. The amplifier is driven by a voltage source with a 250-Ω resistance. What is the input resistance of the amplifier? What is the overall voltage gain G_v ? By what factor must the bias current I_D of the MOSFET be changed so that R_{in} matches R_{sig}?



$R_{in} = 1/g_m = 200\,\Omega$

$V_i = \frac{V_{sig}}{200+250} \cdot 200$, $v_o = g_m v_i (5k \| 5k)$

$G_V = \frac{200}{450} \cdot 12.5 = 5.5\ V/V$

$g_{m_2} = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_{D_2}} = \frac{1}{250} = 4mA/V$

$g_{m_1} = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_{D_1}} = \frac{1}{200} = 5mA/V$

$\Rightarrow \frac{4}{5} = \sqrt{\frac{I_{D2}}{I_{D1}}} \Rightarrow \boxed{\frac{I_{D2}}{I_{D1}} = \frac{16}{25}}$

Q6

7.80 For the circuit in Fig. P7.80, let $R_{sig} \gg r_e$ and $\alpha \simeq 1$. Find v_o.

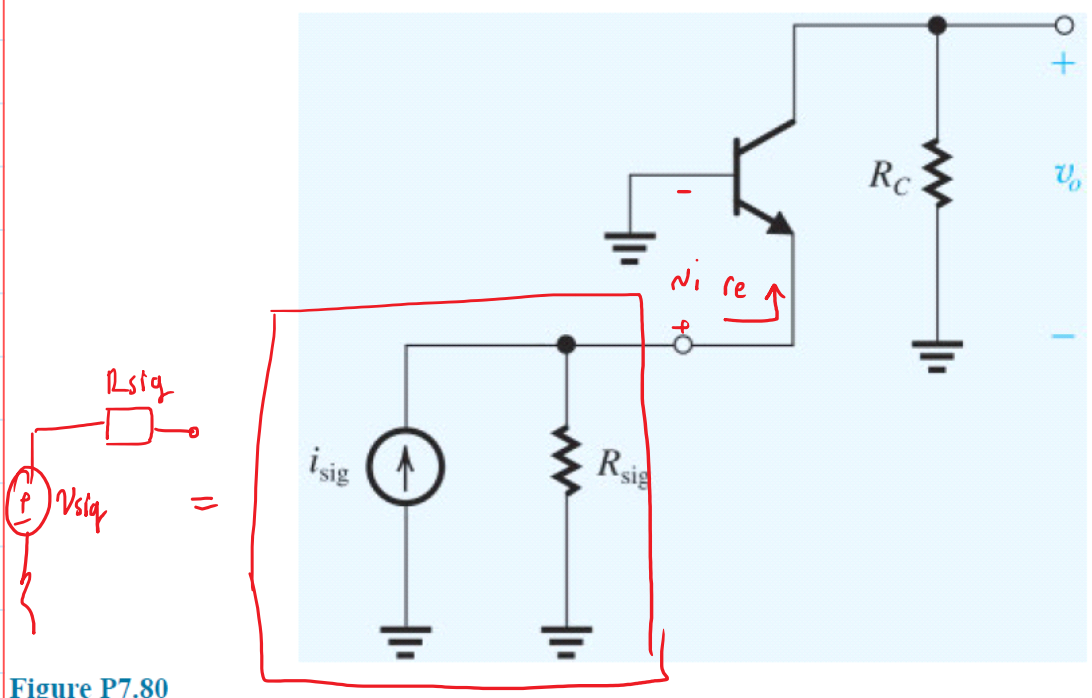


Figure P7.80

$R_{sig} \gg r_e \Rightarrow v_i = -\frac{v_{sig}}{R_{sig}+r_e} \cdot r_e \Rightarrow \frac{v_i}{v_{sig}} \cong \frac{r_e}{R_{sig}}$ where $v_{sig} = i_{sig} R_{sig}$

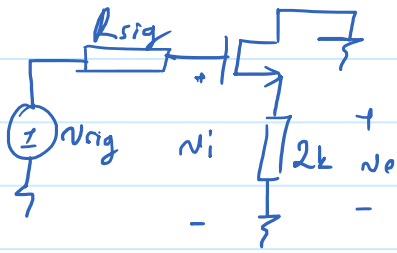
$v_o = g_m R_C v_i = g_m R_C v_{sig} \frac{r_e}{R_{sig}} = \frac{\alpha R_C}{R_{sig}} \cdot i_{sig} R_{sig} = \alpha R_C i_{sig}$

$\alpha \cong 1 \Rightarrow \boxed{v_o = R_C i_{sig}}$

Q7

D 7.83 A source follower is required to deliver a 0.5-V peak sinusoid to a 2- kΩ load. If the peak amplitude of v_{gs} is to be limited to 50 mV, and the MOSFET transconductance parameter k_n is 5 mA/V^2, what is the lowest value of I_D at which the MOSFET can be biased? At this bias current, what are the maximum and minimum currents that the MOSFET will be conducting (at the positive and negative peaks of the output sine wave)? What must the peak amplitude of v_{sig} be?

$$g_m = \sqrt{2 k_n I_D}$$



$$\frac{V_{gs}}{V_o} = \frac{50m}{0.5} = \frac{1}{10} = \frac{r_e}{2k} \Rightarrow r_e = 200\,\Omega$$

$$1/r_e = g_m = 5mA/V = \sqrt{2 k_n I_D} = \sqrt{2.5m I_D}$$

$$I_D = 2.5mA$$

$V_{o\,max} = 0.5V$ sine wave

change in current $\frac{0.5}{2k} = 0.25mA$

$$i_D = I_D \pm i_d = 2.5m \pm 0.25m\ A$$

$$V_{sig_{max}} = \frac{V_{o\,max}}{2k}(2k + r_e) = 0.5 \cdot \frac{22}{2} = 0.55\,V$$

Q8

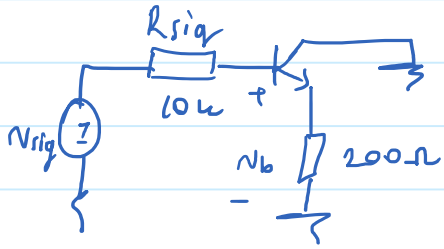
7.85 An emitter follower with a BJT biased at $I_C = 5$ mA and having $\beta = 200$ is connected between a source with $R_{sig} = 10$ kΩ and a load $R_L = 200$ Ω.

(a) Find R_{in}, v_b/v_{sig}, and v_o/v_{sig}.

(b) If the signal amplitude across the base–emitter junction is to be limited to 10 mV, what is the corresponding amplitude of v_{sig} and v_o?

(c) Find the open-circuit voltage gain G_{vo} and the output resistance R_{out}. Use these values first to verify the value of G_v obtained in (a), then to find the value of G_v obtained with R_L reduced to 150 Ω.

a)



$\beta = 200 \Rightarrow r_e = \alpha \frac{V_T}{I_C} \cong \frac{V_T}{I_C} = \frac{25m}{5m} = 5$

$R_{in} = (\beta+1)(r_e + 200) = 201 \cdot 205 \cong 41k\Omega$

$V_b/V_{sig} = \frac{41k}{41k+10k} = \frac{41}{51}$

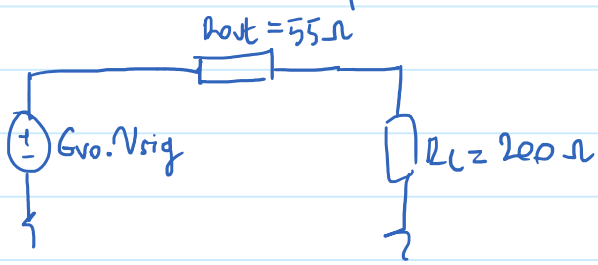
$V_o/V_{sig} = \frac{V_b}{V_{sig}} \frac{V_o}{V_b} = \frac{41}{51} \cdot \frac{200}{200+5}$

$V_o/V_{sig} \cong 0.78$

b) $V_{be} = 10mV$ $\quad V_b = 10m \frac{205}{5} = 410\,mV$ $\quad V_{sig} = 410m \cdot \frac{51}{41} = 510mV$

$V_o = 410m \frac{200}{205} = 400mV$

c) $G_{vo} = 1$, $\quad R_{out} = \frac{R_{sig}}{\beta+1} + r_e = \frac{10k}{201} + 5 \cong 55$



$G_v = G_{vo} \frac{200}{255} \cong 0.78$ V/V

$R_L \to 150$, $\quad G_v = G_{vo} \cdot \frac{150}{150+55} \cong 0.73$ V/V

Q9

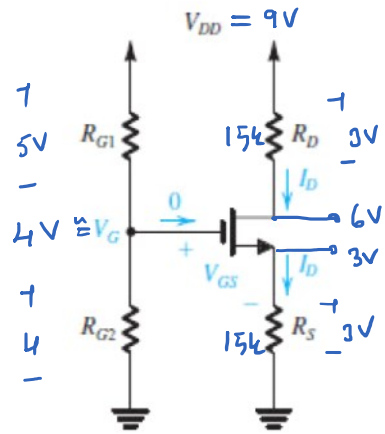
VE 7.2

D 7.92 Consider the classical biasing scheme shown in Fig. 7.50(c), using a 9-V supply. For the MOSFET, $V_t = 0.7$ V, $\lambda = 0$, and $k_n = 5\text{mA/V}^2$. Arrange that the drain current is 0.2 mA, with about one-third of the supply voltage across each of R_S and R_D. Use 22 MΩ for the larger of R_{G1} and R_{G2}. What are the values of R_{G1}, R_{G2}, R_S, and R_D that you have chosen? Specify them to two significant digits. For your design, how far is the drain voltage from the edge of saturation?

$\frac{R_{G1}}{R_{G2}} = \frac{5}{4} = \frac{22M}{R_{G2}}$

$R_{G2} = 17.6M\Omega$

$R_{G1} = 22M\Omega$



(c)

$R_D = R_S = \frac{3}{0.2m} = 15k\Omega$

$I_D = \frac{1}{2} k_n V_{ov}^2 = 0.2m$ (5m)

$V_{ov} = 0.28V$

$V_{GS} = V_{ov} + V_t \cong 1V$

Figure 7.50 (c) Biasing using a fixed voltage at the gate, V_G, and a resistance in the source lead, R_S: practical implementation using a single supply.

$V_{DS} > V_{ov} \Rightarrow V_{DS} = 3V > V_{ov} \cong 0.3$ $\quad V_{DS} - V_{ov} \cong 2.7V$

Q10

D 7.103 Figure P7.103 shows a variation of the feedback-bias circuit of Fig. 7.52. Using a 3-V supply with an NMOS transistor for which $V_t = 0.8$ V, $k_n = 4$ mA/V^2, and $\lambda = 0$, provide a design that biases the transistor at $I_D = 0.5$ mA, with V_{DS} large enough to allow saturation operation for a 1-V negative signal swing at the drain. Use 13 MΩ as the largest resistor in the feedback-bias network. What values of R_D, R_{G1}, and R_{G2} have you chosen?

$V_{DS} - V_{OV} \geq 1$

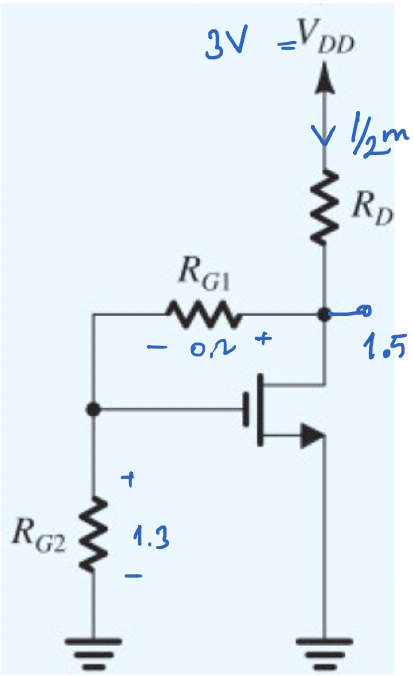
$V_{DS} - V_{GS} + V_t = 1$

$V_D - V_G = 0.2$

$I_D = \frac{1}{2} k_n V_{OV}^2$

$\frac{1}{2}m = \frac{1}{2} 4m\, V_{OV}^2 \Rightarrow V_{OV} = 1/2$

$V_{GS} - V_t = 0.5 \rightarrow V_{GS} = 1.3V$



then $\frac{R_{G1}}{R_{G2}} = \frac{2}{13}$

$R_{G2} = 13M\Omega$

$R_{G1} = 2M\Omega$

$R_D = \frac{3 - 1.5}{1/2m} = 3k\Omega$

Figure P7.103

Q11

D *7.108 You are required to design the bias circuit of Fig. 7.54(a) for a BJT whose nominal β =100.

(a) Find the largest ratio (R_B/R_E) that will guarantee I_E remains within ±5% of its nominal value for β as low as 50 and as high as 150.

(b) If the resistance ratio found in (a) is used, find an expression for the voltage $V_{BB} \equiv V_{CC}R_2/(R_1 + R_2)$ that will result in a voltage drop of $V_{CC}/3$ across R_E.

(c) For V_{CC} =5 V, find the required values of R_1, R_2, and R_E to obtain I_E =0.5 mA and to satisfy the requirement for stability of I_E in (a).

(d) Find R_C so that V_{CE} =1.0 V for β equal to its nominal value.

Check your design by evaluating the resulting range of I_E.

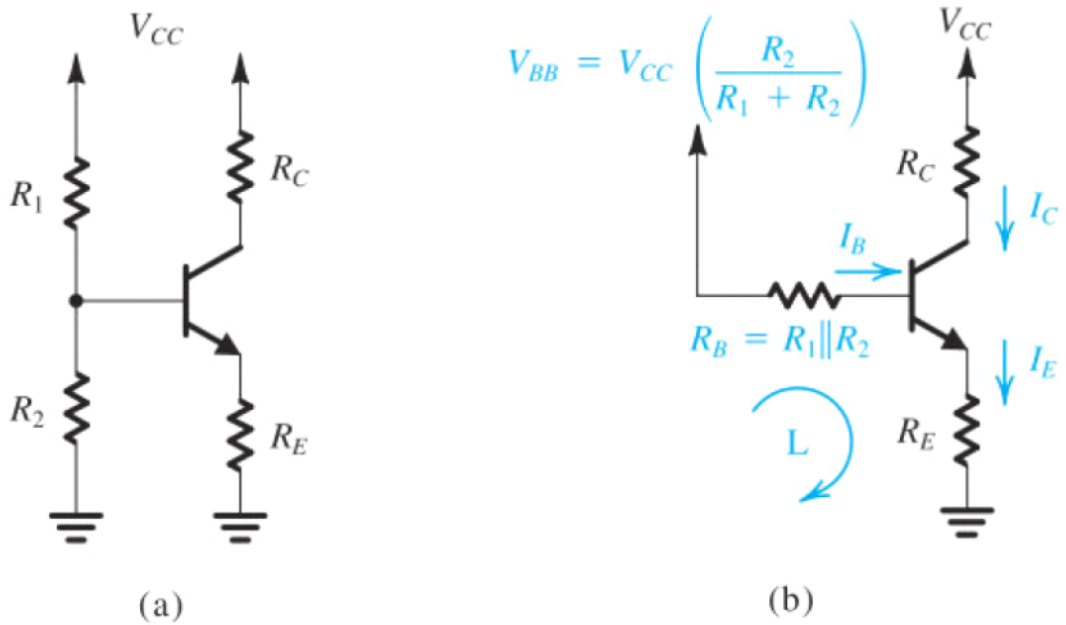


Figure 7.54 Classical biasing for BJTs using a single power supply: (**a**) circuit; (**b**) circuit with the voltage divider supplying the base replaced with its Thévenin equivalent.

(a) $V_{BB} - I_E/\beta+1\ R_B = V_{BE} + I_E R_E \Rightarrow I_E = \dfrac{V_{BB} - V_{BE}}{R_E + R_B/(\beta+1)}$ $\approx$ or

$I_{Enom} = \dfrac{V_{BB} - V_{BE}}{R_E + R_B/101}$ $\quad I_{Emin} = \dfrac{V_{BB} - V_{BE}}{R_E + R_B/51}$ $\quad I_{Emax} = \dfrac{V_{BB} - V_{BE}}{R_E + R_B/151}$

$\dfrac{I_{Emin}}{I_{Enom}} = 0.95 = \dfrac{R_E/R_B + 1/101}{R_E/R_B + 1/51} \Rightarrow \dfrac{R_E}{R_B} = 0.17$

(b) $R_E \cdot I_E = V_{CC}/3$ & $R_E = R_B(0.17) \Rightarrow (0.17) R_B \cdot I_E = V_{CC}/3 \Rightarrow V_{CC} \approx \frac{1}{2} R_B \cdot I_E$

$V_{BB} = V_{CC} R_2/(R_1+R_2) = \frac{1}{2} R_B I_E R_2/(R_1+R_2)$

Q11

D *7.108 You are required to design the bias circuit of Fig. 7.54(a) for a BJT whose nominal β =100.

(a) Find the largest ratio (R_B/R_E) that will guarantee I_E remains within ±5% of its nominal value for β as low as 50 and as high as 150.

(b) If the resistance ratio found in (a) is used, find an expression for the voltage $V_{BB} \equiv V_{CC}R_2/(R_1 + R_2)$ that will result in a voltage drop of $V_{CC}/3$ across R_E.

(c) For V_{CC} =5 V, find the required values of R_1, R_2, and R_E to obtain I_E =0.5 mA and to satisfy the requirement for stability of I_E in (a).

(d) Find R_C so that V_{CE} =1.0 V for β equal to its nominal value.

Check your design by evaluating the resulting range of I_E.

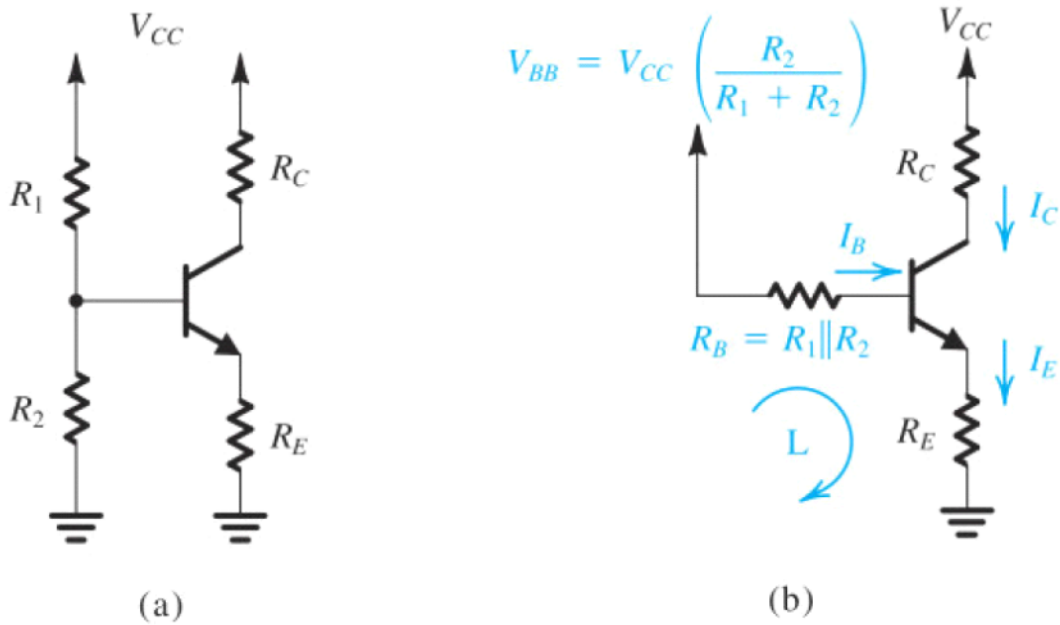


Figure 7.54 Classical biasing for BJTs using a single power supply: (**a**) circuit; (**b**) circuit with the voltage divider supplying the base replaced with its Thévenin equivalent.

(c) $R_E = R_B(0.17)$ $\quad V_{CC} = 5V$ $\quad V_{RE} = 5/3 \Rightarrow R_E = \frac{5/3}{0.5m} = 3.3k$

$R_B = R_E/0.17 = \frac{3.3k}{0.17} \approx 19.4k = \frac{R_1 \cdot R_2}{R_1+R_2}$ $\quad V_{BB} = \frac{V_{CC}}{3} + 0.7 + R_B \cdot \frac{I_E}{101}$

$V_{BB} = 2.13V = 5 \cdot \frac{R_2}{R_1+R_2} = \frac{5 \cdot R_2}{R_1+R_2}(19.4k) \Rightarrow R_1 = 45.5k\Omega, R_2 = 34.6k\Omega$

(d) $V_{CC} \approx V_{CE} + (R_C+R_E) I_E \Rightarrow 5 \approx 1 + (R_C+R_E)(0.5m)$

$R_C + R_E = 8k \Rightarrow R_C = 8k - R_E = 8k - 3.3k = 4.7k\Omega$

Q12

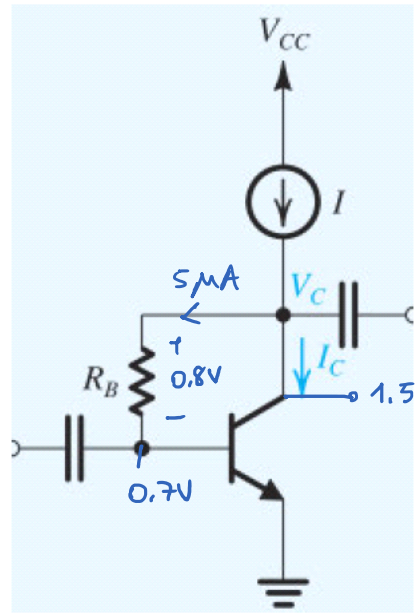


Figure P7.113

D 7.113 A circuit that can provide a very large voltage gain for a high-resistance load is shown in Fig. P7.113. Find the values of I and R_B to bias the BJT at $I_C = 0.5$ mA and $V_C = 1.5$ V. Let $\beta = 100$.

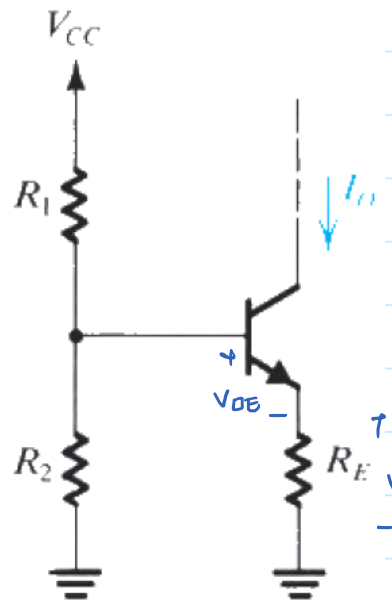
$I_B = I_c/\beta = 0.5m/100 = 5\mu A$ $\quad R_B = \frac{0.8}{5\mu} = 160k\Omega$

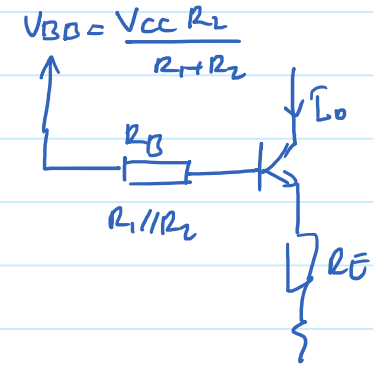
$I = I_B + I_c = 0.505mA \simeq 0.5mA$

Q13

7.114 The circuit in Fig. P7.114 provides a constant current I_O as long as the circuit to which the collector is connected maintains the BJT in the active mode. Show that

$$I_O = \alpha \frac{V_{CC}[R_2/(R_1+R_2)] - V_{BE}}{R_E + (R_1 \| R_2)/(\beta+1)}$$





$V_E = R_E \cdot I_E$

Figure P7.114

$$V_{BB} = \frac{I_O}{\beta} \cdot R_1 // R_2 + V_{BE} + \frac{I_O}{\alpha} \cdot R_E$$

$$I_O = \alpha \frac{V_{BB} - V_{BE}}{R_E + \alpha \cdot R_1 // R_2 / \beta} = \alpha \frac{V_{CC}(R_2/R_1+R_2) - V_{BE}}{R_E + (R_1 // R_2)/(\beta+1)}$$

Q14

VE 7.3

D 7.129 In the circuit of Fig. P7.129, v_{sig} is a small sine-wave signal with zero average. The transistor β is 100.

(a) Find the value of R_E to establish a dc emitter current of about 0.2 mA.

(b) Find R_C to establish a dc collector voltage of about +0.5 V.

(c) For R_L = 10 kΩ, draw the small-signal equivalent circuit of the amplifier and determine its overall voltage gain.

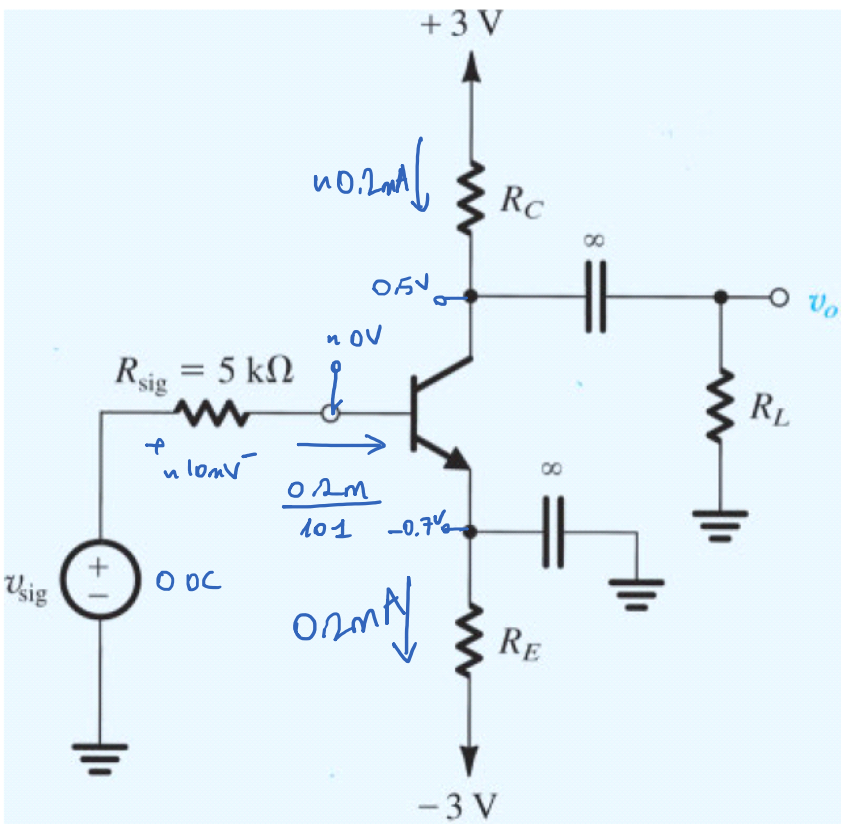


Figure P7.129

(a) $R_E \approx \frac{-0.7+3}{0.2m} = \frac{2.3}{1/5m} = 11.5\,k\Omega$

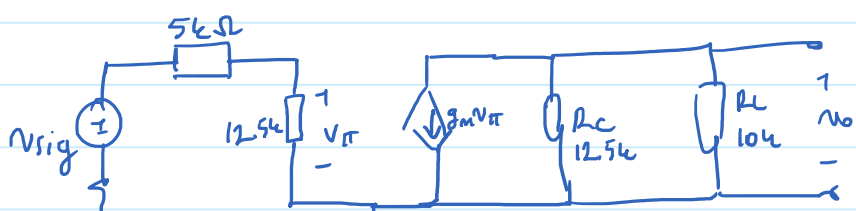
(b) $R_C \approx \frac{3-0.5}{0.2m} = \frac{2.5}{1/5m} = 12.5\,k\Omega$

$r_e = V_T/I_E = 25m/0.2m = 125\,\Omega$

$r_\pi = (\beta+1)r_e \approx 12.5\,k\Omega$

$g_m = I_C/V_T \approx \frac{0.2m}{0.025} \approx 8\,mA/V$

(c) $R_L = 10\,k\Omega$



$G_v = \frac{12.5}{17.5} \cdot (-8m) \cdot (12.5k \,//\, 10k) = 31.7\,V/V$

Q15

***7.130** The amplifier of Fig. P7.130 consists of two identical common-emitter amplifiers connected in cascade. Observe that the input resistance of the second stage, R_{in2}, constitutes the load resistance of the first stage.

(a) For $V_{CC} = 15$ V, $R_1 = 100$ kΩ, $R_2 = 47$ kΩ, $R_E = 3.9$ kΩ, $R_C = 6.8$ kΩ, and $\beta = 100$, determine the dc collector current and dc collector voltage of each transistor.

(b) Draw the small-signal equivalent circuit of the entire amplifier and give the values of all its components.

(c) Find R_{in1} and v_{b1}/v_{sig} for $R_{sig} = 5$ kΩ.

(d) Find R_{in2} and v_{b2}/v_{b1}.

(e) For $R_L = 2$ kΩ, find v_o/v_{b2}.

(f) Find the overall voltage gain v_o/v_{sig}.

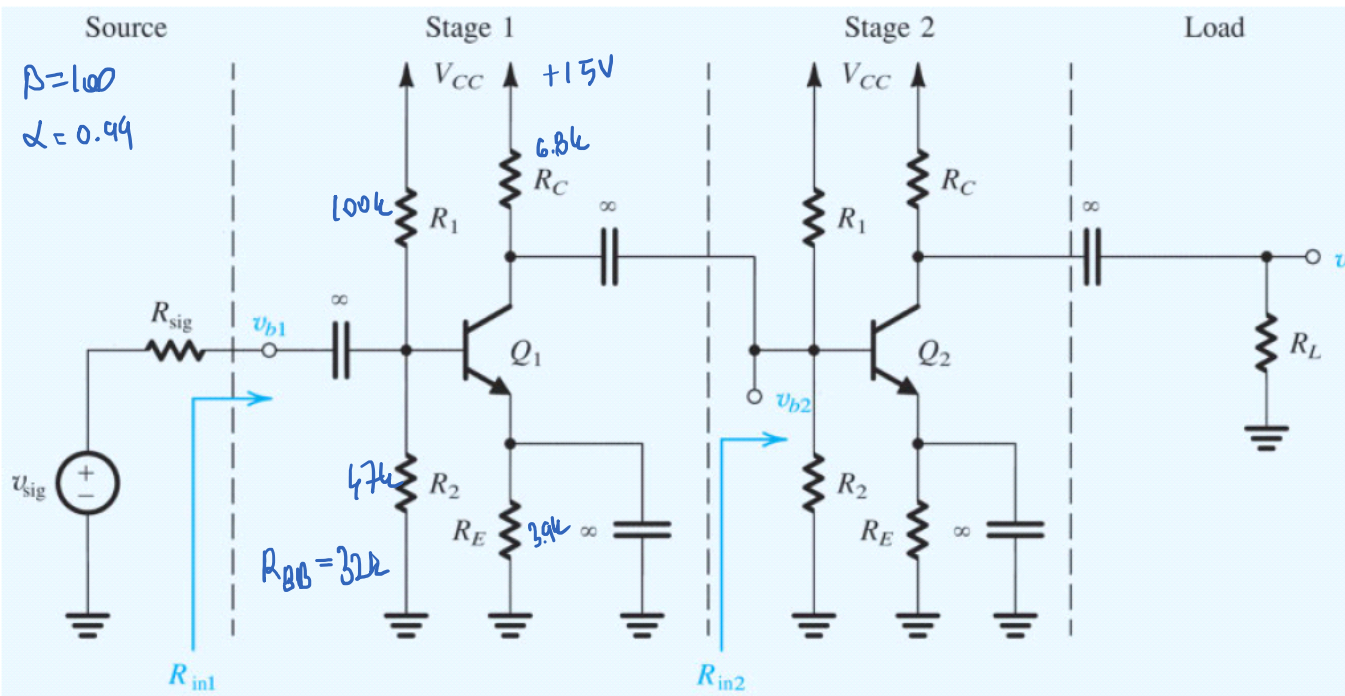
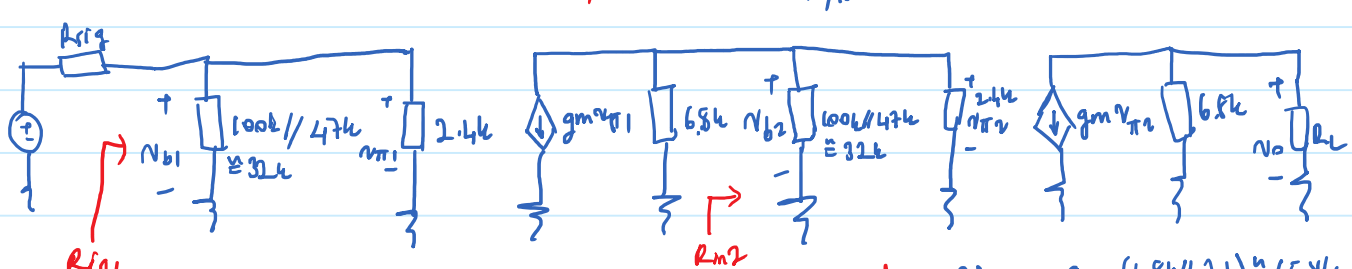


Figure P7.130

(a) $V_{BB} \cong \frac{15 \cdot 47}{147} \cong 4.8\text{V}$ $V_E \cong 4.8 - 0.7 = 4.1\text{V}$ $I_E = \frac{4.1}{3.9k} = 1.05\text{mA} \Rightarrow I_C = 1.04\text{mA}$

$V_C = 15 - (6.8k)(1.04m) = 7.07\text{V}$ $r_\pi = \frac{V_T}{I_B} \cong \frac{25m}{1.04m/100} \cong 2.4k\Omega$ $g_m = \frac{I_C}{V_T} = \frac{1.04m}{0.025} \cong 42\text{mA/V}$

(b)



(c) $R_{in1} = 100k // 47k // 2.4k \cong 2.23k$ $\frac{v_{b1}}{v_{sig}} = \frac{2.23k}{5k + 2.23k} \cong 0.31$

(d) $R_{in2} = 2.23k\Omega$ $v_{b2}/v_{b1} = -g_m(100k // 47k // 6.8k // 2.4k) \cong 68\text{ V/V}$

(e) $\frac{v_o}{v_{b2}} = -g_m(6.8k // 2k) \cong 65\text{ V/V}$

(f) $\frac{v_o}{v_{sig}} = \frac{v_{b1}}{v_{sig}} \cdot \frac{v_{b2}}{v_{b1}} \cdot \frac{v_o}{v_{b2}} \cong 1370\text{ V/V}$

Q16

VE 7.4

7.131 In the circuit of Fig. P7.131, the BJT is biased with a constant-current source, and v_{sig} is a small sine-wave signal. Find R_{in} and the gain v_o/v_{sig}. Assume $\beta = 100$. If the amplitude of the signal v_{be} is to be limited to 5 mV, what is the largest signal at the input? What is the corresponding signal at the output?

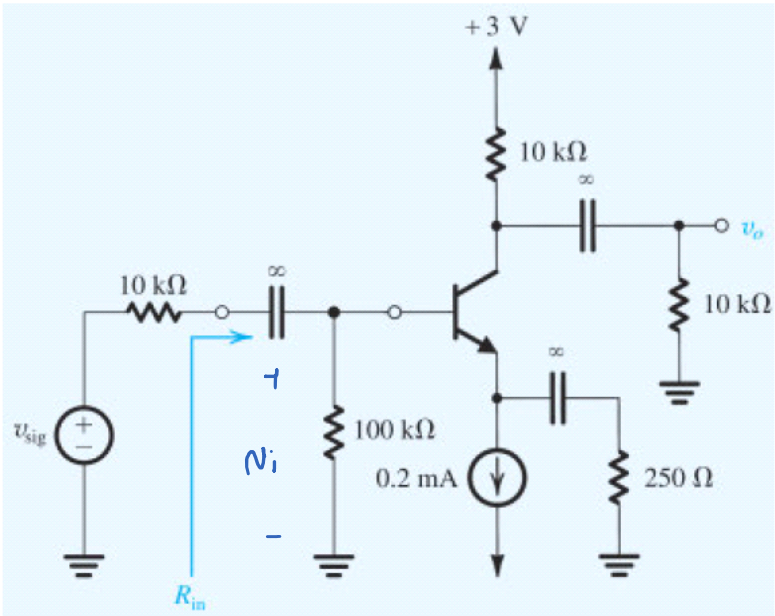


Figure P7.131

$I_E = 0.2mA \Rightarrow \alpha = 0.99 \Rightarrow I_C \cong 0.2mA$

$g_m = \frac{0.2m}{0.025} = 8mA/V \quad , \quad r_e = \frac{\alpha}{g_m} \cong 125\Omega$

$R_{in} = 100k \,//\, (\beta+1)(r_e + 250) \cong 100k \,//\, 38k \cong 28k$

$G_V = \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \cdot \frac{v_o}{v_i} = \frac{R_i}{R_i + R_{sig}} \cdot \frac{R_C // R_L}{r_e + R_e} = \frac{28k}{28k+10k} \cdot \frac{10k // 10k}{125+250} \cong 9.8 \, V/V$

$v_{be} = \frac{v_i}{250+125} \cdot 125 = v_i/3 \qquad v_{be} = 5mV \Rightarrow v_i = 15mV \Rightarrow v_o = G_V \cdot v_i = (9.8)(15m)$

$\max v_o \cong 147mV$

Q17

7.132 The BJT in the circuit of Fig. P7.132 has $\beta = 100$.

(a) Find the dc collector current and the dc voltage at the collector.

(b) Replacing the transistor with its T model, draw the small-signal equivalent circuit of the amplifier. Analyze the resulting circuit to determine the voltage gain v_o/v_i.

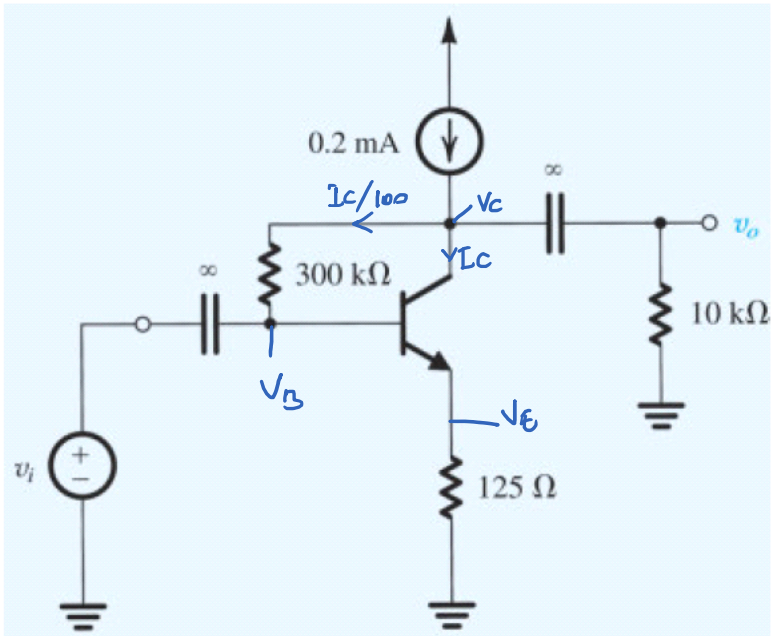


Figure P7.132

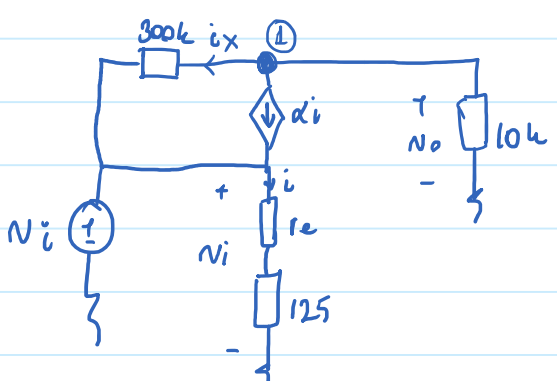
$I_C\left(1+1/100\right) = 0.2mA \Rightarrow I_C \cong 0.2\,mA$

$V_{CB} = 300k.\,I_B \cong 300k.\frac{0.2m}{100} = 0.6V$ $\quad V_B = V_{BE} + 125(0.2m) \cong 0.7 + 0.25$

$V_B = 0.95$

$V_C = V_{CB} + V_B = 0.95 + 0.6 = 1.55V$

$re = \frac{V_T}{I_E} = \frac{25m}{0.2m} = 125\Omega$, $gm = \frac{\Delta}{re} \cong 8mA/V$



$i = \frac{Ni}{Re+125} = \frac{Ni}{250}$

$No = -(\alpha i + ix).10k$

$No = -\left(0.99\frac{Ni}{250} + \frac{No - Ni}{300k}\right)10k \cong -40Ni - No/30$

$No\left(1+1/30\right) = -40Ni \Rightarrow No/Ni \cong -39\,V/V$

Q18

D 7.133 For the circuit in Fig. P7.133, find the value of the bias current I that results in $R_{in} = R_{sig}$. Also, find the voltage gain v_o/v_{sig}. Assume that the source provides a small signal v_{sig} and that $\beta = 100$.

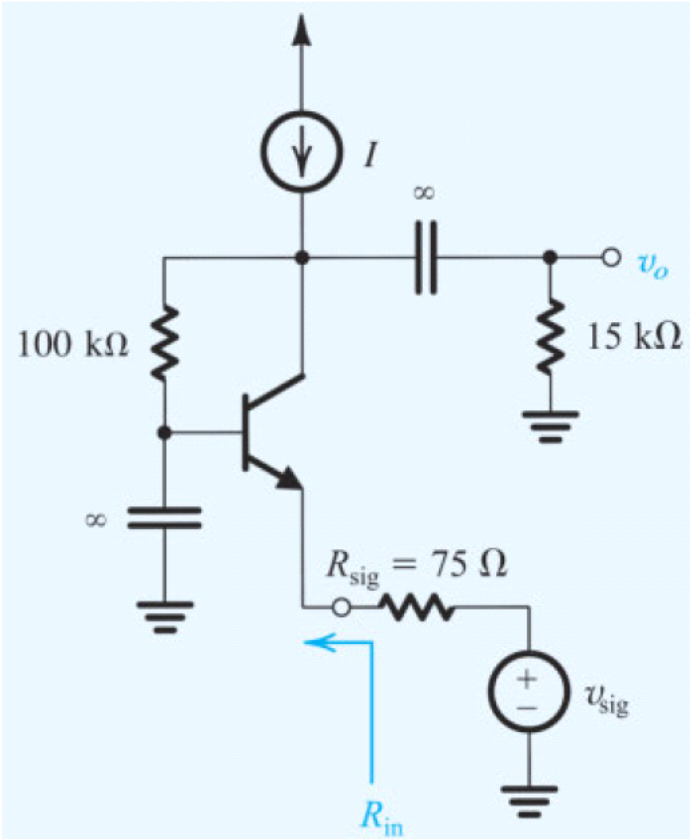
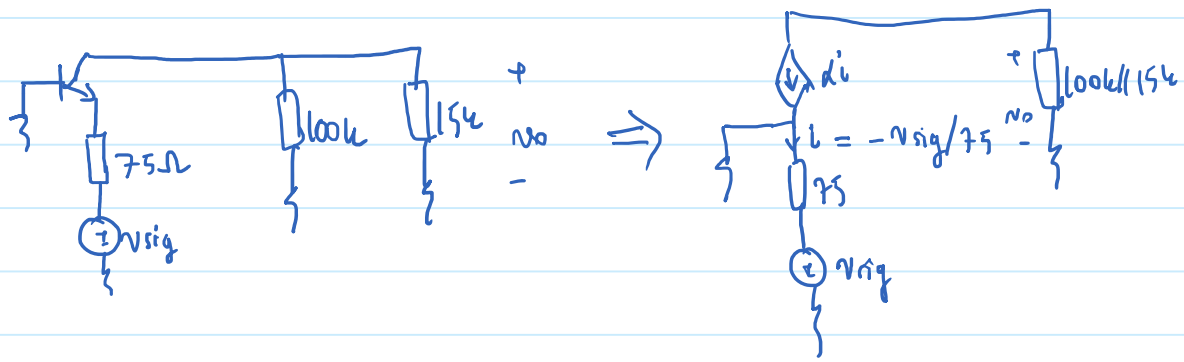


Figure P7.133

$R_{sig} = 75 = r_e = \frac{V_T}{I_E} \Rightarrow I_E = \frac{25m}{75} = 1/3\,mA = I$



$V_o = -\alpha i \cdot (100k // 15k) = -(0.99)(-V_{sig}/75)(100k // 15k) = 172\,V_{sig}$

$V_o/V_{sig} = 172\ V/V$

Q19

7.134 For the emitter-follower circuit shown in Fig. P7.134, the BJT used is specified to have β values in the range of 80 to 180 (a distressing situation for the circuit designer).

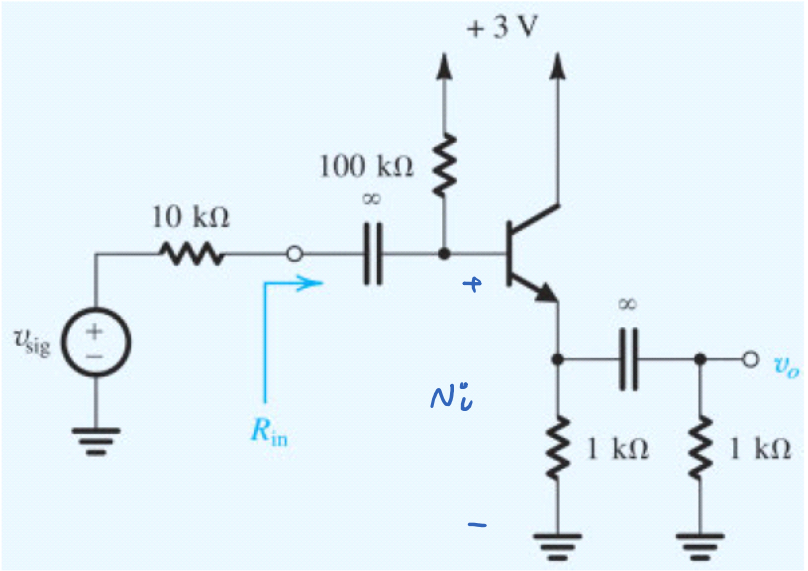


Figure P7.134

For the two extreme values of β (β =80 and β =180), find:

(a) I_E, V_E, and V_B

(b) the input resistance R_{in}

(c) the voltage gain v_o/v_{sig}

(a) $V_{CC} - 100k\, I_E/\beta+1 = V_{BE} + I_E.(1k)$

$3 - 0.7 \cong I_E\,(1k + 100k/\beta+1) \Rightarrow I_E = \dfrac{2.3}{1k + \frac{100k}{\beta+1}}$

$\beta=80 \Rightarrow I_E \cong 1.03\,mA$

$V_E = I_E.1k = 1.03\,V$

$V_B \cong V_E + 0.7 = 1.73\,V$

$r_e = \dfrac{V_T}{I_E} = \dfrac{25m}{1.03m} \cong 24\,\Omega$

$\beta = 180 \Rightarrow I_E \cong 1.48\,mA$

$V_E \cong I_E.1k = 1.48\,V$

$V_B \cong V_E + 0.7 = 2.18\,V$

$r_e = \dfrac{V_T}{I_E} = \dfrac{25m}{1.48m} \cong 17\,\Omega$

(b) $R_{in} = 100k \,//\, (\beta+1)(r_e + \underbrace{R_e}_{12k//12k}) = 100k \,//\, (\beta+1)(r_e + 500)$

$\beta=80 \Rightarrow R_{in} \cong 30k\,\Omega$, $\beta=180 \Rightarrow R_{in} \cong 48k\,\Omega$

(c) $\dfrac{v_o}{v_{sig}} = \dfrac{R_{in}}{R_{in}+10k} \cdot \dfrac{500}{r_e+500} \Rightarrow \beta=80 \Rightarrow v_o/v_{sig} \cong 0.72$, $\beta=180 \Rightarrow v_o/v_{sig} \cong 0.8$

Q20

7.135 For the emitter follower in Fig. P7.135, the signal source is directly coupled to the transistor base. If the dc component of v_{sig} is zero, find the dc emitter current. Assume β =100. Neglecting r_o, find R_{in}, the voltage gain v_o/v_{sig}, the current gain i_o/i_i, and the output resistance R_{out}.

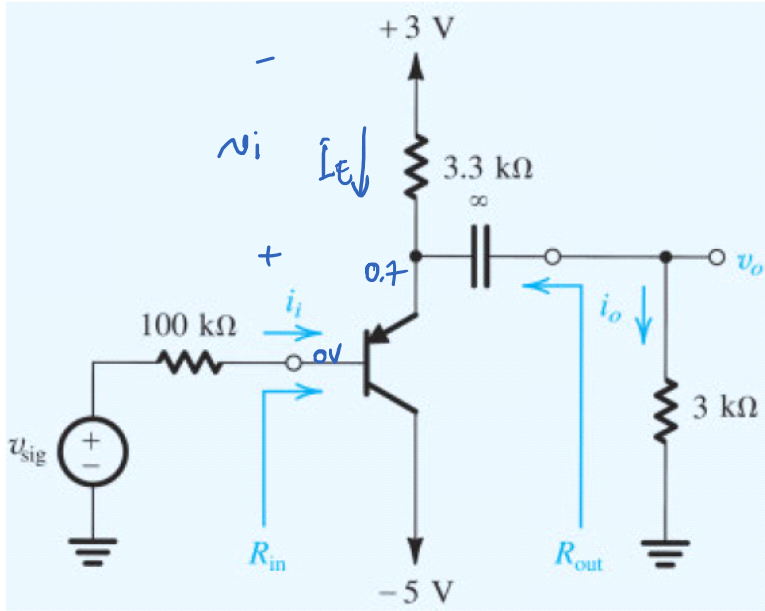


Figure P7.135

$$I_E = \frac{3-0.7}{3.3k} \approx 0.7mA \qquad r_e = \frac{V_T}{I_E} \approx 36\Omega$$

$$R_{in} = \left(r_e + (3.3k \,//\, 3k)\right)(\beta+1) \approx 162k\Omega$$

$$v_o/v_{sig} = \frac{v_i}{v_{sig}} \cdot \frac{v_o}{v_i} = \frac{162k}{162k+100k} \cdot \frac{3.3k\,//\,3k}{3.3k\,//\,3k + r_e} \approx 0.62\ V/V$$

$$i_o/i_i = \frac{v_o/3k}{v_{sig}/100k + R_{in}} = \frac{v_o}{v_{sig}} \cdot \frac{262k}{3k} \approx 54\ A/A$$

$$R_{out} = \left(\frac{100k}{\beta+1} + r_e\right) \,//\, (3.3k) \approx 780\Omega$$